# Practical 1 - Revisiting Pandas and NumPy

## Problem Statement

The objective of this practical is to understand the fundamentals of the Pandas and NumPy libraries in Python. The practical involves creating raw student data, converting it into a Pandas DataFrame, analyzing NumPy arrays by determining their shape, size, and dimensions, performing basic matrix operations such as addition, subtraction, multiplication, transpose, inverse, and matrix product, and converting categorical data into numerical data using One-Hot Encoding.

## Solution

### Step 1: Importing Libraries, Creating Raw Student Data and Converting it into a Pandas DataFrame

In this step, the required Python libraries, Pandas and NumPy, are imported. A raw dataset containing information of ten students is created using a Python dictionary. The dataset includes Student ID, Name, Age, Gender, City, Department, and Marks.
The raw data is then converted into a Pandas DataFrame, which organizes the information into a structured tabular format. Finally, the basic details of the DataFrame, such as its shape, column names, and data types, are displayed.

### Step 2: Creating NumPy Arrays and Performing Matrix Operations

In this step, two NumPy matrices are created to perform basic numerical computations. The properties of both matrices, including their shape, size, and dimensions, are determined. Various matrix operations such as addition, subtraction, element-wise multiplication, matrix product, transpose, and inverse are then performed to understand the capabilities of the NumPy library for numerical and matrix-based computations.

### Step 3: Applying One-Hot Encoding

In this step, the categorical columns namely Gender, City, and Department are converted into numerical format using the One-Hot Encoding technique provided by the Pandas library. Each unique category is represented as a separate binary column containing values of 0 and 1.

This preprocessing technique is widely used in Data Science and Machine Learning because most algorithms require numerical input instead of categorical text values.

## Complete Python Code

```
# ============================================================
# Practical 1 - Revisiting Pandas and NumPy
# MSc Data Science
# ============================================================


# ------------------------------------------------------------
# Step 1 : Import Required Libraries
# ------------------------------------------------------------

import pandas as pd
import numpy as np

# ------------------------------------------------------------
# Step 2 : Creating Raw Student Data
# ------------------------------------------------------------

raw_student_data = {
    "Student_ID": [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],

    "Name": [
        "Rahul",
        "Priya",
        "Amit",
        "Neha",
        "Karan",
        "Sneha",
        "Rohan",
        "Anjali",
```

```
        "Vikas",
        "Pooja"
    ],

    "Age": [
        22, 23, 21, 22, 24,
        23, 22, 21, 24, 23
    ],

    "Gender": [
        "Male",
        "Female",
        "Male",
        "Female",
        "Male",
        "Female",
        "Male",
        "Female",
        "Male",
        "Female"
    ],

    "City": [
        "Delhi",
        "Mumbai",
        "Shimla",
        "Delhi",
        "Chandigarh",
        "Mumbai",
        "Delhi",
        "Shimla",
        "Chandigarh",
```

```
        "Delhi"
    ],

    "Department": [
        "Data Science",
        "Data Science",
        "Artificial Intelligence",
        "Artificial Intelligence",
        "Cyber Security",
        "Cyber Security",
        "Data Science",
        "Artificial Intelligence",
        "Cyber Security",
        "Data Science"
    ],

    "Marks": [
        88,
        91,
print(student_df.dtypes)

print("=" * 70)

# ============================================================
# Step 5 : Creating NumPy Arrays
# ============================================================

matrix_A = np.array([
    [2, 4],
    [6, 8]
])
```

```
matrix_B = n        75,
        85,
        79,
        90,
        83,
        87,
        81,
        95
    ]
}

# ------------------------------------------------------------
# Step 3 : Convert Raw Data into Pandas DataFrame
# ------------------------------------------------------------

student_df = pd.DataFrame(raw_student_data)

# ------------------------------------------------------------
# Step 4 : Display the DataFrame
# ------------------------------------------------------------

print("=" * 70)
print("STUDENT DATAFRAME")
print("=" * 70)

print(student_df)

# ------------------------------------------------------------
# Basic Information about the DataFrame
# ------------------------------------------------------------

print("\nNumber of Rows and Columns:")
```

```
print(student_df.shape)

print("\nColumn Names:")
print(student_df.columns)

print("\nData Types:")
p.array([
    [1, 3],
    [5, 7]
])

print("\n" + "=" * 70)
print("NUMPY MATRICES")
print("=" * 70)

print("\nMatrix A:")
print(matrix_A)

print("\nMatrix B:")
print(matrix_B)

# ============================================================
# Step 6 : Shape, Size and Dimension
# ============================================================

print("\n" + "=" * 70)
print("PROPERTIES OF MATRIX A")
print("=" * 70)

print("Shape      :", matrix_A.shape)
print("Size       :", matrix_A.size)
print("Dimension  :", matrix_A.ndim)
```

```python
print("\n" + "=" * 70)
print("PROPERTIES OF MATRIX B")
print("=" * 70)

print("Shape      :", matrix_B.shape)
print("Size       :", matrix_B.size)
print("Dimension  :", matrix_B.ndim)

# ============================================================
# Step 7 : Matrix Operations
# ============================================================

print("\n" + "=" * 70)
print("MATRIX OPERATIONS")
print("=" * 70)

# Addition
print("\n1. Addition (A + B)")
addition = matrix_A + matrix_B
print(addition)

# Subtraction
print("\n2. Subtraction (A - B)")
subtraction = matrix_A - matrix_B
print(subtraction)

# Element-wise Multiplication
print("\n3. Element-wise Multiplication (A * B)")
multiplication = matrix_A * matrix_B
print(multiplication)
```

```python
# Matrix Product
print("\n4. Matrix Product (A × B)")
matrix_product = np.dot(matrix_A, matrix_B)
print(matrix_product)

# Transpose
print("\n5. Transpose of Matrix A")
transpose = matrix_A.T
print(transpose)

# Inverse
print("\n6. Inverse of Matrix A")
inverse = np.linalg.inv(matrix_A)
print(inverse)



# ============================================================
# Step 8 : One-Hot Encoding
# ============================================================

print("\n" + "=" * 70)
print("ONE-HOT ENCODING")
print("=" * 70)

# Selecting categorical columns
categorical_columns = ["Gender", "City", "Department"]

# Performing One-Hot Encoding
encoded_df = pd.get_dummies(
    student_df,
    columns=categorical_columns,
    dtype=int
```

```
)

print("\nOriginal DataFrame:")
print(student_df)

print("\nDataFrame After One-Hot Encoding:")
print(encoded_df)
```

# Output

## Output 1: Student DataFrame

```
================================================================================
anjanibharatchauhan@MacBook-Air Practical 1 - Revisiting Pandas and NumPy: % python3 practical1.py
================================================================================
STUDENT DATAFRAME
================================================================================
   Student_ID    Name  Age  Gender        City              Department  Marks
0         101   Rahul   22    Male       Delhi            Data Science     88
1         102   Priya   23  Female      Mumbai            Data Science     91
2         103    Amit   21    Male      Shimla  Artificial Intelligence     75
3         104    Neha   22  Female       Delhi  Artificial Intelligence     85
4         105   Karan   24    Male  Chandigarh          Cyber Security     79
5         106   Sneha   23  Female      Mumbai          Cyber Security     90
6         107   Rohan   22    Male       Delhi            Data Science     83
7         108  Anjali   21  Female      Shimla  Artificial Intelligence     87
8         109   Vikas   24    Male  Chandigarh          Cyber Security     81
9         110   Pooja   23  Female       Delhi            Data Science     95
```

## Output 2: DataFrame Information

```
Number of Rows and Columns:
(10, 7)

Column Names:
Index(['Student_ID', 'Name', 'Age', 'Gender', 'City', 'Department', 'Marks'], dtype='str')

Data Types:
Student_ID    int64
Name            str
Age           int64
Gender          str
City            str
Department      str
Marks         int64
dtype: object
================================================================================
```

## Output 3: NumPy Matrices

```
================================================================================
NUMPY MATRICES
================================================================================

Matrix A:
[[2 4]
 [6 8]]

Matrix B:
[[1 3]
 [5 7]]
```

## Output 4: Shape, Size and Dimension

```
================================================================================
PROPERTIES OF MATRIX A
================================================================================
Shape      : (2, 2)
Size       : 4
Dimension  : 2

================================================================================
PROPERTIES OF MATRIX B
================================================================================
Shape      : (2, 2)
Size       : 4
Dimension  : 2
```

## Output 5: Matrix Operations

```
================================================================================
MATRIX OPERATIONS
================================================================================

1. Addition (A + B)
[[ 3  7]
 [11 15]]

2. Subtraction (A - B)
[[1 1]
 [1 1]]

3. Element-wise Multiplication (A * B)
[[ 2 12]
 [30 56]]

4. Matrix Product (A × B)
[[22 34]
 [46 74]]

5. Transpose of Matrix A
[[2 6]
 [4 8]]

6. Inverse of Matrix A
[[-1.    0.5 ]
 [ 0.75 -0.25]]
```

## Output 6: One-Hot Encoding

```
============================================================
ONE-HOT ENCODING
============================================================

Original DataFrame:
   Student_ID    Name  Age  Gender       City              Department  Marks
0         101   Rahul   22    Male      Delhi            Data Science     88
1         102   Priya   23  Female     Mumbai            Data Science     91
2         103    Amit   21    Male     Shimla  Artificial Intelligence     75
3         104    Neha   22  Female      Delhi  Artificial Intelligence     85
4         105   Karan   24    Male  Chandigarh          Cyber Security     79
5         106   Sneha   23  Female     Mumbai          Cyber Security     90
6         107   Rohan   22    Male      Delhi            Data Science     83
7         108  Anjali   21  Female     Shimla  Artificial Intelligence     87
8         109   Vikas   24    Male  Chandigarh          Cyber Security     81
9         110   Pooja   23  Female      Delhi            Data Science     95

DataFrame After One-Hot Encoding:
   Student_ID    Name  Age  Marks  ...  City_Shimla  Department_Artificial Intelligence  Department_Cyber Security  Department_Data Science
0         101   Rahul   22     88  ...            0                                   0                          0                        1
1         102   Priya   23     91  ...            0                                   0                          0                        1
2         103    Amit   21     75  ...            1                                   1                          0                        0
3         104    Neha   22     85  ...            0                                   1                          0                        0
4         105   Karan   24     79  ...            0                                   0                          1                        0
5         106   Sneha   23     90  ...            0                                   0                          1                        0
6         107   Rohan   22     83  ...            0                                   0                          0                        1
7         108  Anjali   21     87  ...            1                                   1                          0                        0
8         109   Vikas   24     81  ...            0                                   0                          1                        0
9         110   Pooja   23     95  ...            0                                   0                          0                        1

[10 rows x 13 columns]
```

## Conclusion

In this practical, the basic concepts of the Pandas and NumPy libraries were successfully explored. Raw student data was created using a Python dictionary and converted into a Pandas DataFrame for structured data analysis. NumPy was used to create matrices and perform operations such as finding shape, size, dimensions, addition, subtraction, multiplication, matrix product, transpose, and inverse. Finally, categorical variables were converted into numerical format using One-Hot Encoding. This practical helped in understanding the fundamentals of data manipulation and numerical computation, which are essential concepts in Data Science and Machine Learning.


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